

Sample IC Visual Inspection Report

Fictional training document for PDF parsing, chunking, retrieval, and citation testing.

Document ID	SAMPLE-RAG-IC-001
Device Under Test	QFN 48 Mixed-Signal Controller
Lot ID	TRAINING-LOT-042
Date	2026-06-17
Prepared By	Demo QA Engineering Team
Confidentiality	Public sample - no customer or company data

1. Objective

The objective of this inspection was to evaluate visible package quality, marking consistency, lead condition, and surface anomalies for a sample batch of QFN 48 integrated circuits. The inspection workflow combined manual review guidelines with AI assisted image analysis. This document is intentionally fictional and is designed to provide searchable content for a RAG backend.

2. Inspection Scope

The inspection covered five areas: package body condition, top side marking, logo alignment, pinhole like surface defects, and lead row geometry. The process did not include electrical testing, decapsulation, X-ray inspection, or chemical analysis. All conclusions in this sample report are based only on visual inspection images.

Inspection Area	Method	Acceptance Criteria
Package body	Image quality review and edge boundary check	No major crack, chip-out, or deformation
Top marking	OCR comparison against expected part number	Part number must match expected text
Logo	Template similarity comparison	Similarity score should be above 0.85
Pinhole-like defects	Quadrant-based defect counting	No quadrant should contain excessive concentrated defects
Lead rows	Row detection and spacing consistency check	No missing, bridged, or severely bent leads

3. Data Inputs

The input dataset contained 120 top side device images captured under controlled lighting. Images were stored as PNG files and grouped by sample tray position. Each image filename included a tray identifier, row number, column number, and capture timestamp. The expected marking text for the device was QFN48-MS-2601.

Example filename format:

TRAY02_R04_C07_20260617_143022.png

4. AI Assisted Workflow

The workflow first checked whether the package body was visible and centered. A detection model then identified the package boundary, top marking region, logo region, and four lead row regions. Optical character recognition was applied to the marking region. A texture comparison method was used to highlight unusual surface patterns. Pinhole-like candidates were counted separately for the upper-left, upper-right, lower-left, and lower-right

quadrants.

The AI model was not used as the final decision maker. Its role was to prioritize images for human review, standardize measurement outputs, and reduce repetitive manual screening. Final pass/fail decisions remained subject to human quality review.

5. Retrieval-Friendly Key Facts

This section is intentionally written with clear facts so that a RAG system can retrieve them during testing.

Question Type	Answerable Fact
Expected marking	The expected marking text is QFN48-MS-C-2601.
Number of images	The dataset contains 120 top side device images.
Inspection methods	The workflow uses package detection, OCR, logo similarity, texture comparison, and quadrant-based pinhole counting.
Final decision	The AI model does not make the final decision; human quality review makes the final pass/fail call.
Out-of-scope items	Electrical testing, decapsulation, X-ray inspection, and chemical analysis are out of scope.

6. Observations

Most devices showed consistent package edges and readable markings. Seven images were flagged for manual review because the OCR confidence was below 0.70. Three images showed possible smudging near the marking region. Two images showed lead row spacing irregularity on the lower edge. No severe package cracking was observed in the sample batch.

Logo similarity scores ranged from 0.86 to 0.94, with a median score of 0.91. Because the minimum score remained above the review threshold of 0.85, no logo mismatch was reported. Pinhole-like surface candidates were detected in 14 images, but the candidates were sparse and not concentrated in a single quadrant.

Metric	Result	Interpretation
OCR low confidence count	7 images	Manual review required
Possible marking smudge	3 images	Minor cosmetic concern
Lead-row spacing irregularity	2 images	Review lower edge images
Severe package crack	0 images	No severe package damage observed
Median logo similarity	0.91	Above threshold
Pinhole-like candidates	14 images	Sparse distribution

7. Preliminary Disposition

The preliminary disposition for this fictional lot is conditional pass pending manual review of the seven OCR low confidence images and the two lower edge lead row irregularity images. No evidence in this sample report supports a severe package-level rejection. The recommended next action is to review the flagged image list and document whether each flagged case is a true defect or an imaging artifact.

8. Recommended Follow-Up

The quality reviewer should inspect the flagged OCR images at full resolution, compare the observed marking against the expected text QFN48-MSC-2601, and determine whether the low OCR confidence is caused by real marking degradation or by glare. For lead row issues, the reviewer should compare lower edge lead spacing against neighboring devices from the same tray.

For future automation work, the team should add more labeled examples of glare, partial smudge, and benign surface texture variation. These examples can improve model robustness and reduce false positive review flags.

9. Example RAG Test Questions

Test Question	Expected Source Area
What is the expected marking text?	Section 3 or Section 5
How many images were inspected?	Section 3 or Section 5
Which inspection methods were used?	Section 2, Section 4, or Section 5
Did the AI model make the final pass/fail decision?	Section 4 or Section 5
How many images had low OCR confidence?	Section 6
What is the preliminary disposition?	Section 7

10. Disclaimer

This report is a synthetic sample created for software testing. It does not represent a real customer, real lot, real inspection result, or proprietary standard. It may be used freely as a local test PDF for document upload, chunking, vector retrieval, and source citation development.